

AYUSH AGARWAL

Vellore Institute of Technology

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Education

• Vellore Institute of Technology B.Tech in Computer Science Specialization in Data Science	Sep 2022 – Present Vellore, Tamil Nadu	CGPA 8.99
• Intermediate Class 12 th	July 2020 – July 2022 Stepping Stones, Chandigarh	77%
• Class 10 th	2020 The Gurukul, Panchkula, Haryana	89%

Experience (Internship)

Panacea Infosec PVT LTD IT Intern	June 2024 – July 2024 Noida, UP
- PCI DSS Knowledge: Understanding standards and requirements. - Compliance Support: Assist with documentation, audits, risk assessments, and scans. - Risk Assessment: Identify and evaluate payment data security risks. Technical Skills:- Network Security: Firewalls and segmentation, Encryption: Secure data in transit and storage, Compliance & Auditing: Monitor activity and automate checks.	

Projects

Toxic Comment Classifier Chat Forum Flask, TensorFlow, Keras, LSTM, GloVe, PyMongo, MongoDB 	Mar 2025
• Developed a deep learning-based toxic comment classifier using LSTM and GloVe embeddings, integrated into a Flask-powered Q&A forum. • Automatically detects and blocks toxic questions or replies in real-time, showing user warnings before submission. • Designed and implemented a responsive chat interface and connected it to MongoDB using PyMongo.	
Retrieval-Augmented Generation (RAG) Chatbot LangChain, Python, FAISS/ChromaDB, LLMs	July 2026
• Built a RAG chatbot that retrieves information from PDF and text documents using LangChain and an LLM. • Implemented document loading, text chunking, and preprocessing to improve retrieval accuracy. • Generated embeddings and stored them in FAISS/ChromaDB for efficient semantic search. • Integrated a retrieval pipeline with conversational memory for context-aware multi-turn responses.	
Integrated Algorithmic Trading Framework Backtrader + Binance 	Jan 2026
• Developed an end-to-end algorithmic trading framework integrating Backtrader with Binance API for seamless backtesting and live/paper trading. • Implemented and compared Simple Linear, AR(3), and PyTorch-based Neural Network models; AR(3) achieved best performance with Sharpe Ratio 2.28. • Built a complete trading pipeline including OHLC data processing, SMA/EMA/RSI feature engineering, automated signal generation, and realistic cost-adjusted evaluation (78.98% gross return under 3× leverage).	
Telecom Fraud Detection System Tenserflow, Python, Deep Learning, Streamlit 	Mar 2026
• Developed an end-to-end Telecom Fraud Detection system using Deep Learning (TensorFlow) achieving ~83% accuracy for binary classification of fraudulent call patterns. • Performed data preprocessing, feature engineering, and model evaluation (train-test split) to optimize fraud prediction performance. • Built a Streamlit-based real-time ML web application with explainable AI module to generate interpretable fraud detection insights.	
Autism Spectrum Disorder (ASD) Prediction Model Python, Scikit-learn, ML, Streamlit 	Sep 2024
• Developed an ML model to predict Autism Spectrum Disorder from behavioral and demographic data. • Preprocessed data (normalization, encoding, outlier handling) and applied supervised algorithms (Logistic Regression, SVM, Decision Trees), evaluating with accuracy, ROC-AUC, and F1-score.	

Certifications

- **GITHUB Link:** <https://github.com/ayushagarwal120904-star>
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- Machine Learning, Data Science and Generative AI with Python
- Deep Learning 2025: Neural Networks & AI
- AI Automation in n8n: Build Agents & Voice Agents
- AI Engineer Agentic Track: The Complete Agent & MCP Course
- AI Engineer Core Track: LLM Engineering, RAG, QLORA, Agents

Technical Skills

Languages: C++, Python, C, HTML/CSS, JavaScript, SQL, R

Technologies: Jupyter, MS Visual Studio, R Studio, Office 365 Tools, GitHub

Interpersonal: Communication, Problem-Solving, Networking, Analytical Skills, Strategic Thinking, Teamwork

Coursework: Operating System, Computer Networks, Database Management System, Machine Learning

Solved 1200+ DSA problems on various coding platforms (Leetcode, Codechef, etc), demonstrating strong problem-solving, algorithmic thinking, and optimization skill

Cleared TCS iON NQT – IT with 87.78% overall score and 94.19% in Programming (C++), May 2026.